

DENG Mobility Project

Zurich Pipeline: Mobility and Weather Data

End-to-End Batch Data Pipeline -> PostgreSQL, Airflow, Terraform, GCS and BigQuery

Final Project

Git-Hub Repository: https://github.com/konakus/DENG_Mobility.git

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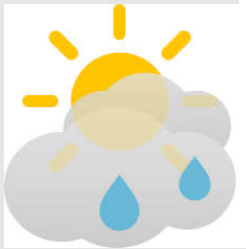
1. Use Case

An urban data analyst from Zurich wants to understand how weather conditions affect mobility.

Key question: How do mobility patterns change depending on rain, temperature, weekdays, or weekends?

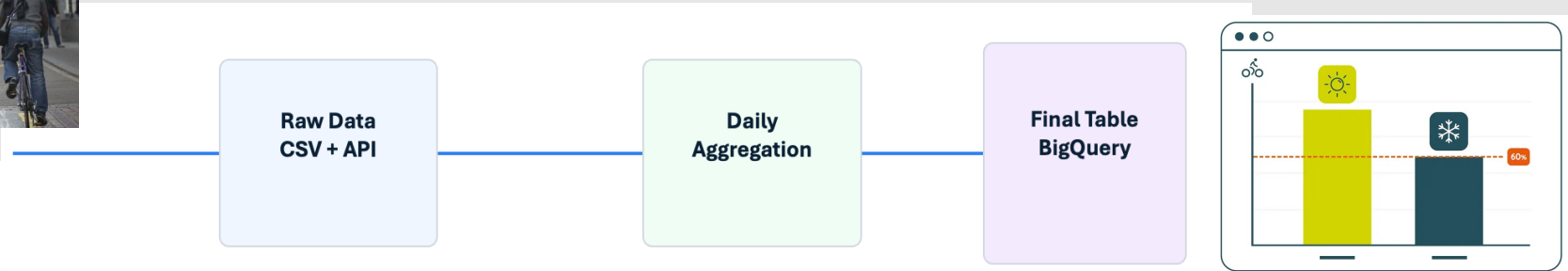
Traffic

Zurich mobility database



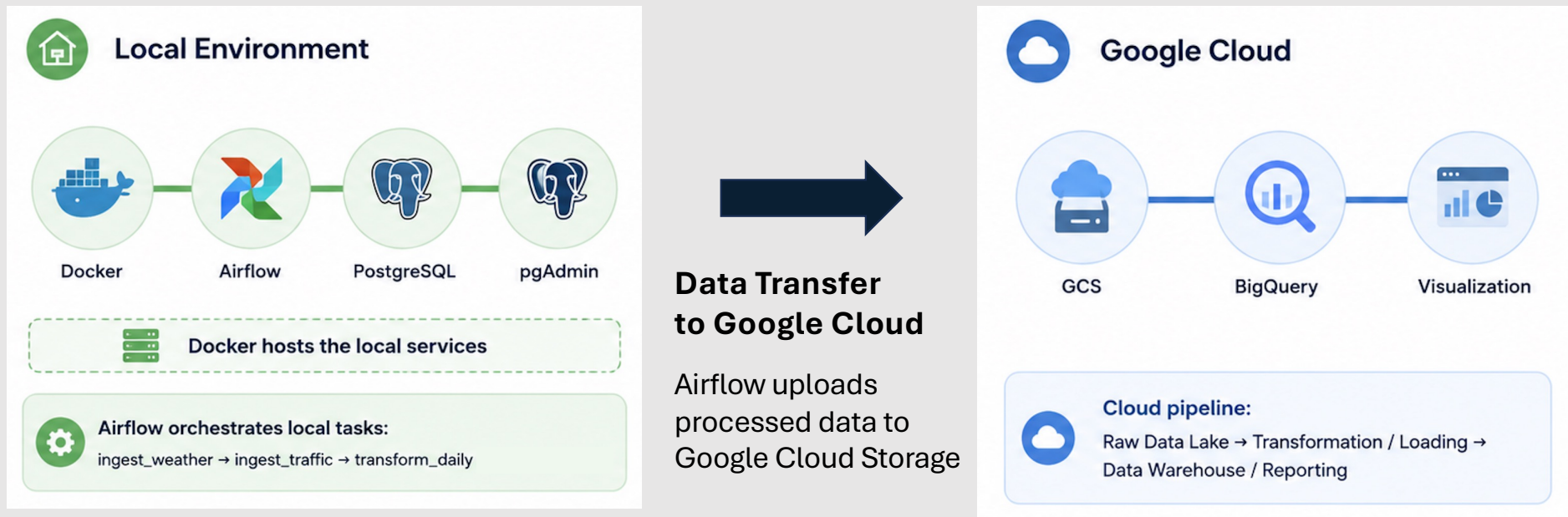
Weather

Open Meteo API



2. Project Architecture

The project combines local development with a cloud pipeline.



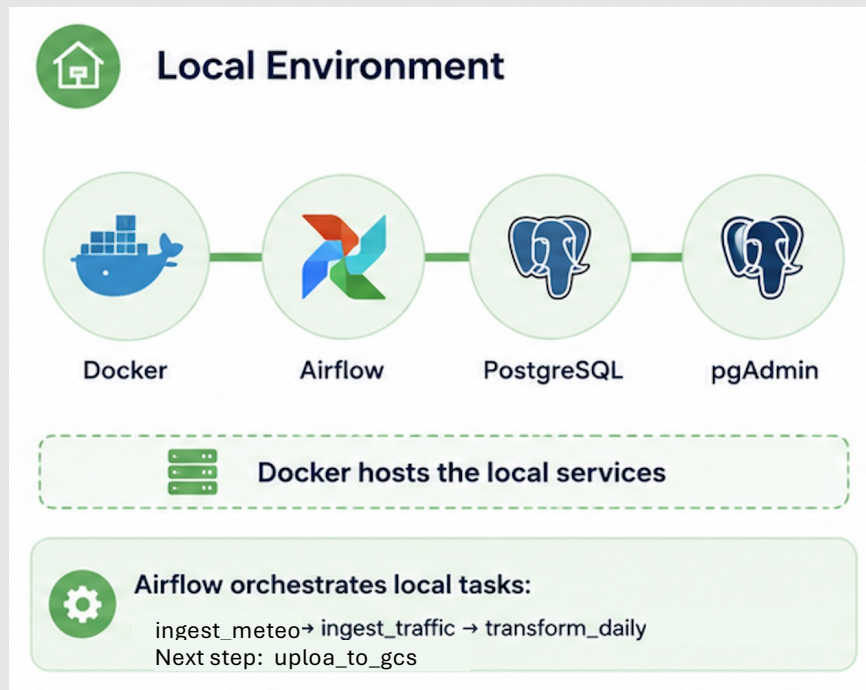
Infrastructure as Code



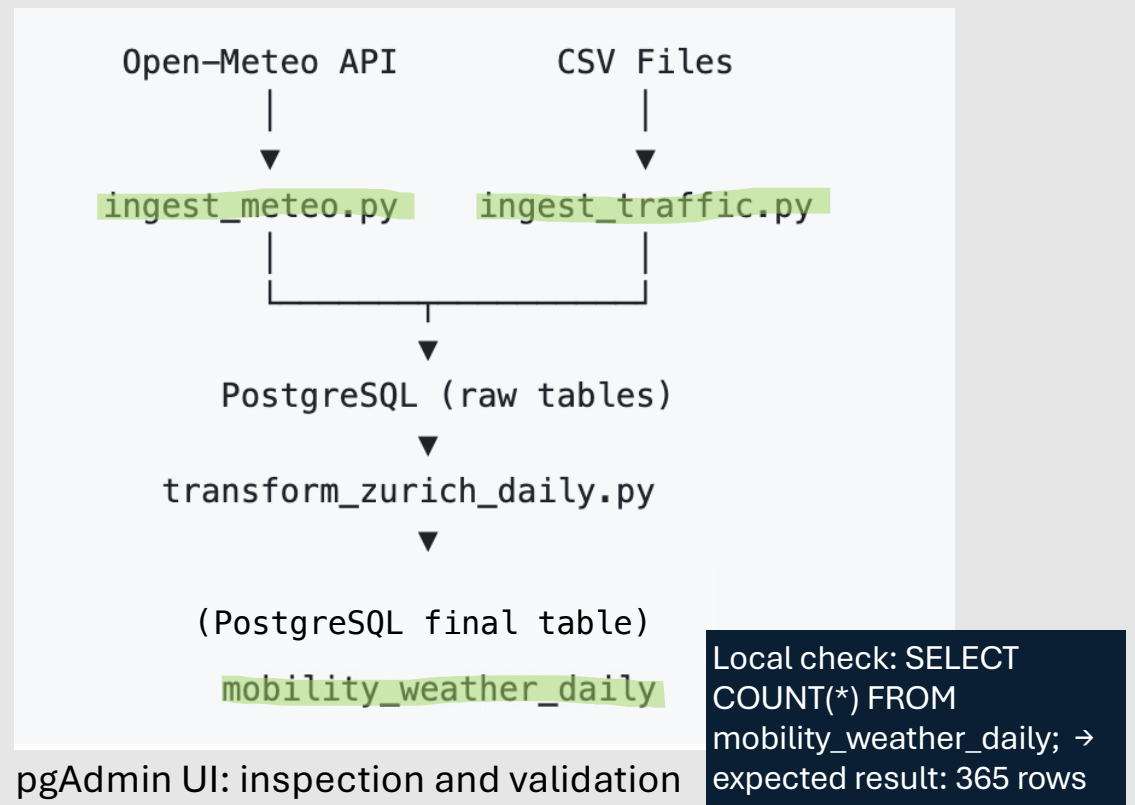
Terraform provisions and manages cloud resources such as GCS buckets, BigQuery datasets, and permissions. Terraform (Infrastructure as Code).

3. Local Pipeline

The project combines local development with a cloud pipeline.



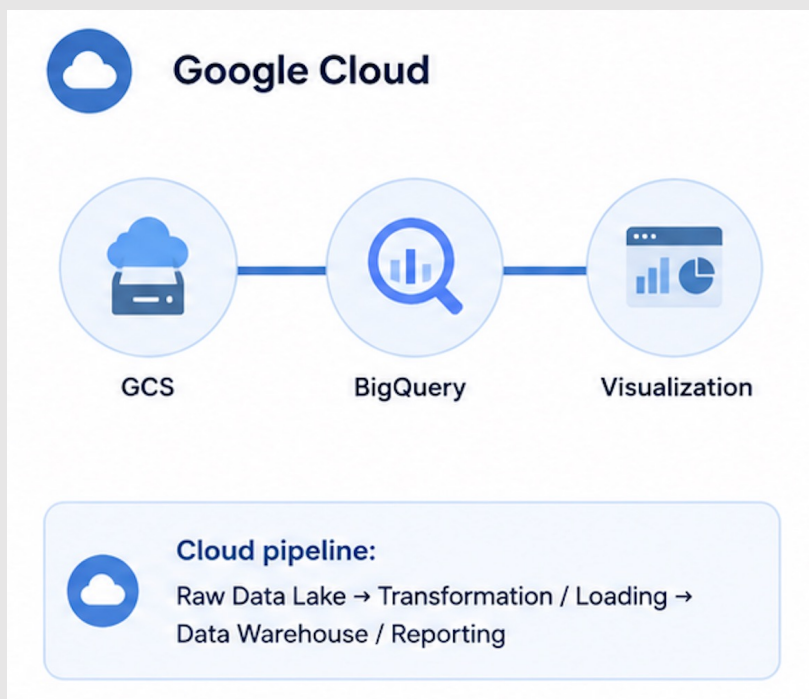
Orchestrated by Airflow



4. Cloud Pipeline - Overview

Airflow transfers data to Google Cloud Storage (GCS)

→ GCS acts as the raw data lake, while BigQuery transforms the data into the final warehouse table.



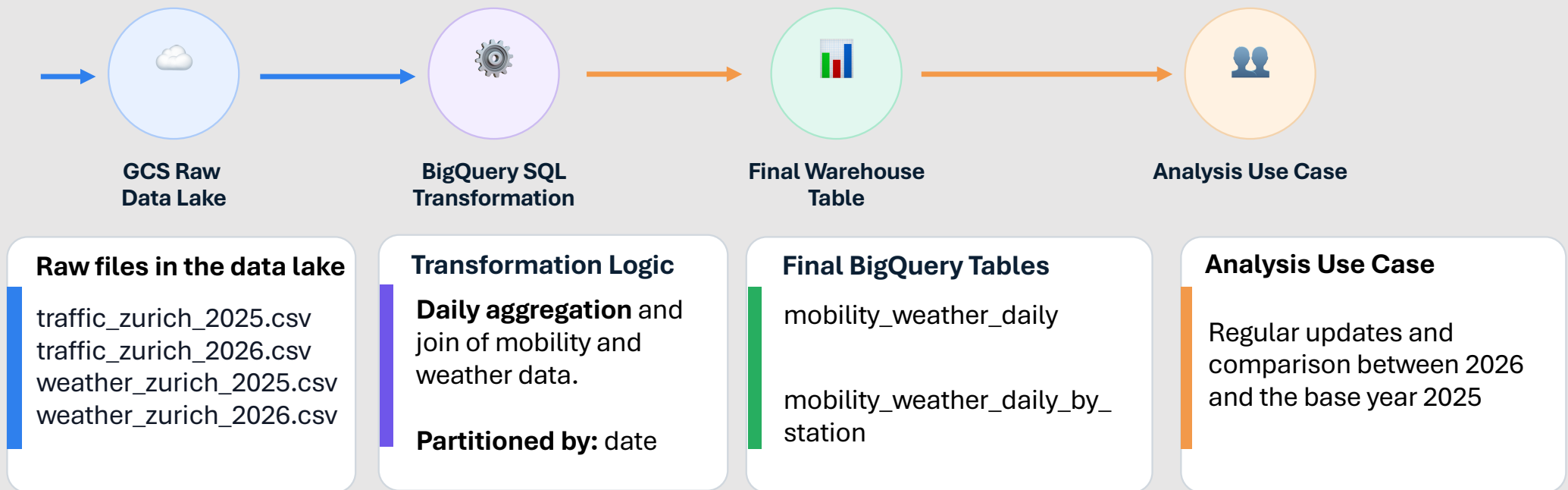
The cloud pipeline consists of **two** Airflow DAGs:

1. Data Lake Ingestion
Source data → GCS
2. Warehouse Transformation
GCS → BigQuery final table

Terraform provisions the GCS bucket and BigQuery dataset used by this pipeline.

5. Cloud Resources and Warehouse Transformation

Raw data is stored in GCS, transformed in BigQuery, and served as a final analysis table.



Terraform creates the cloud resources; Airflow uses them to run the data pipeline.

6. Terraform and Cloud Credentials

Terraform creates the required Google Cloud resources as code.



Terraform creates

- GCS bucket as raw data lake
- BigQuery dataset as data warehouse
- Required IAM permissions



Project resources

Project: projectmobile-494518

Bucket: project-mobile-zurich-data-lake-494518

Dataset:
zurich_mobility_warehouse



Credentials

Service account key is stored locally:
terraform/keys/my-creds.json

Never upload credentials to GitHub. Credentials are excluded via .gitignore.

terraform init → terraform fmt → terraform validate → terraform plan → terraform apply

7. BigQuery Table and Optimization

The final table is ready for analytical queries.

mobility_weather_daily

date

year / month

is_weekend

temperature

precipitation

wind_speed

bicycle_count

pedestrian_count

Partitioning

by date

Good for time filters and time comparisons.

Clustering

City-Level:
year, month, is_weekend

Station-Level:
Station_ID, year, month

Examples of questions

- How do 2025 and 2026 differ?
- What happens on rainy days?
- Are weekends different from weekdays?

9. Verification and Reproducibility

The project clearly demonstrates how results can be reviewed and re-executed.

PostgreSQL

Lokaler Check:
COUNT(*) = 365 rows
für 2025

Cloud Storage

Check whether all raw files are
present in the bucket.

BigQuery

COUNT(*), MIN(date), MAX(date)
check and preview the table.

**Reproducible: git clone → Store credentials locally → Build Docker image
→ docker compose up → Run Airflow DAGs → Check BigQuery**

Important: Google Cloud Keys and Terraform State are not committed to GitHub.

→ Future development: comparison with other cities

10. Dashboard and First Analysis in Looker Studio

After loading the final table into BigQuery, we connected the data to Looker Studio/Data Studio and created a simple dashboard to explore mobility patterns.

The screenshot shows the Google Cloud ProjectMobile interface. The left sidebar contains a list of queries, with '1_Precipitation_Bicycle_Pedest...' selected (marked with a red circle 1). The main panel displays the SQL query (marked with a red circle 2) and the query results (marked with a red circle 3). The query results table is as follows:

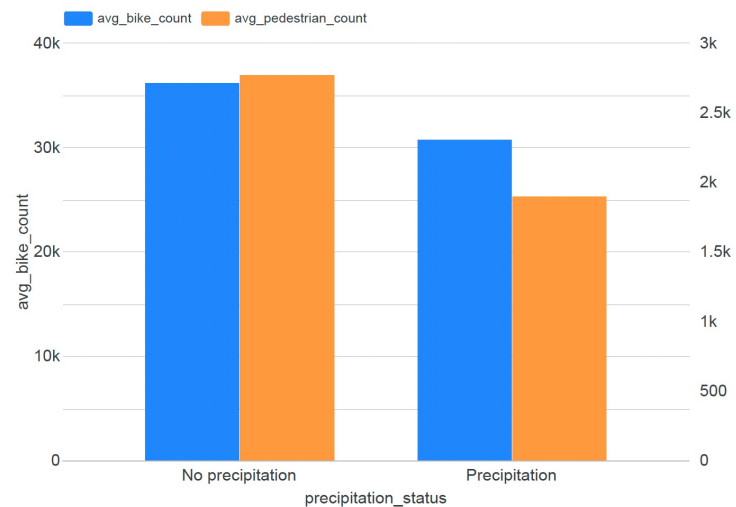
Row	precipitation_status	avg_bike_count	avg_pedestrian_count	total_bike_count	total_pedestrian_count	number_of_days
1	No precipitation	36093.23553719...	2764.223140495...	8734563	668942	
2	Precipitation	30662.04119850...	1894.760299625...	8186765	505901	

The 'Open in' dropdown menu is open, showing options: 'Connected sheets', 'Data Studio' (marked with a red circle 4), 'Notebook', and 'Data canvas'.

Dashboard Analysis: Weather and Weekly Mobility Patterns

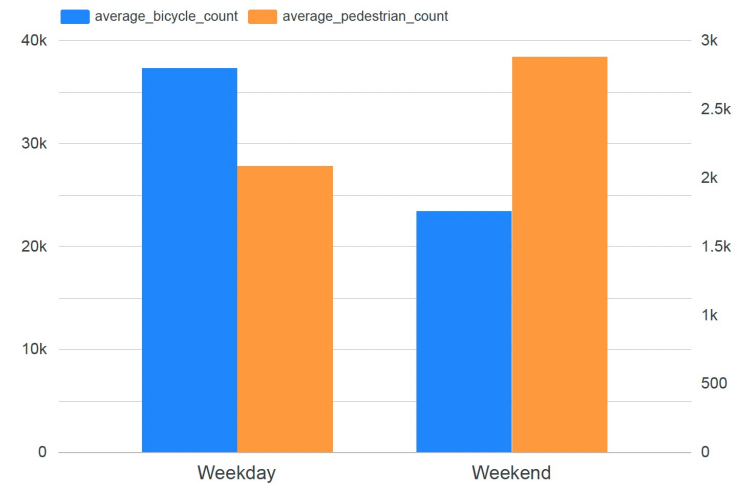
Impact of Precipitation on Bicycle and Pedestrian Traffic

City Zurich: Average daily counts for 2025 and 2026



Average Mobility: Weekday vs. Weekend

City Zurich: Average daily counts for 2025 and 2026

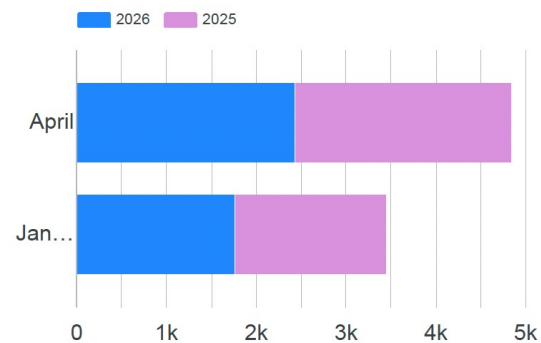


Dashboard Analysis: City vs. Outer District Traffic

Daily Traffic: City vs. Outer District

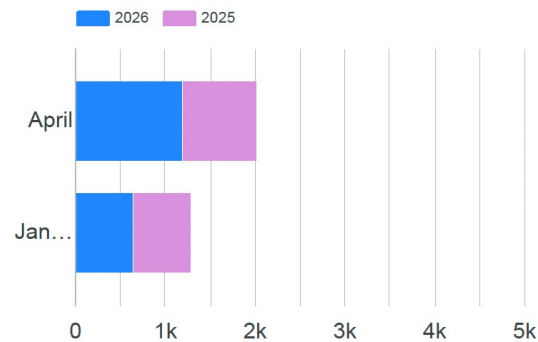
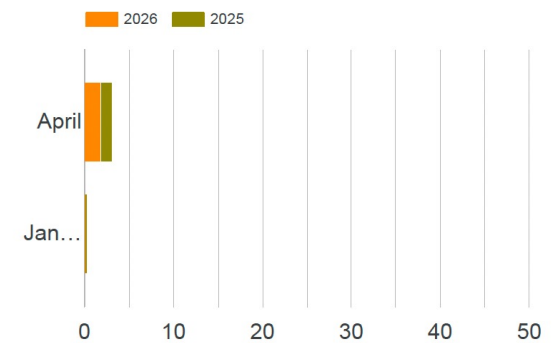
2025 vs. 2026 and January vs. April

Bycicle



City

Pedestrian



Outer
district

